Microservice Architecture (/index.html)

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Pattern: Server-side page fragment composition

pattern (/tags/pattern)

Context

You have applied the Microservice architecture pattern (../microservices.html). Services are developed by business capability (../decomposition/decompose-by-business-capability.html)/subdomain (../decomposition/decompose-by-subdomain.html)-oriented teams that are also responsible for the user experience. Some UI screens/pages display data from multiple service. Consider, for example, an Amazon-style product detail page, which displays numerous data items including:

- Basic information about the book such as title, author, price, etc.
- Your purchase history for the book
- Availability
- Buying options
- Other items that are frequently bought with this book
- Other items bought by customers who bought this book
- Customer reviews
- Sellers ranking
- …

Each data item corresponds to a separate service and how it is displayed is, therefore, the responsibility of a different team.

Problem

How to implement a UI screen or page that displays data from multiple services?

Forces

Solution

Each team developers a web application that generates the HTML fragment that implements the region of the page for their service. A UI team is responsible for developing the page templates that build pages by performing server-side aggregation (e.g. server-side include style mechanism) of the service-specific HTML fragments.

Related patterns

- The Client-side UI composition (client-side-ui-composition.html) pattern is an alternative approach

pattern (/tags/pattern)

Modernization Help (https://chrisrichardson.net/)

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Vikash
6 years ago

Hello. What is a good way to learn more about this pattern? The book written by Chris does not have this chapter yet I believe. thanks

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C **Chuck Zheng**
7 years ago edited

eBay has extend this pattern and API gateway to explicitly compose page from modules on the server-side to minimize client/service roundtrips. And serve the contents to both mobile native apps as well as mobile web apps, while desktop web may have some different modules. We also identified many modules that are reused across many pages, e.g. Marketing module, Ads module, merchanding module, etc - for them we actually let those domain's agile teams to build module providers for integration with hosting page's Experience Services - this allows us to both providing consistent customer experience as well as reducing duplicated engineering cost. here is a blog blog about it

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R **Ram R** → Chuck Zheng
7 years ago

These module providers - do they run on their own instances as microservices themselves? Isnt that a single point of failure in that if such providers fail, the app itself fails though other microservices may be running?

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C **Chuck Zheng** → Ram R
7 years ago

Yes, these module providers can run on their own instances. Using usual HA setup pattern the single point of failure can be removed/reduced - they